ALGEBRAICITY OF THE MODULI STACK OF RELATIVELY PERFECT UNIVERSALLY GLUABLE COMPLEXES

THE STACKS PROJECT AUTHORS

Editorial note. Only the source-defined universally gluable relatively perfect objects are parameterized; this is not a moduli stack of all arbitrary complexes.

Editorial note. The sheaf-moduli special cases are not counted as additional tasks from the same chain.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Derived categories of algebraic spaces, relative perfection and negativeExt vanishing, formal deformation theory and Artin representability. The local stack/diagonal/limits/Rim–Schlessinger/tangent/effectiveness chain is retained author proof rather than granted as the main conclusion.

Source reference. Original section 0DKM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17017–17017. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 0DKN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17018–17018. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08H4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10767–10767. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DKT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17023–17023. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0AA7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12975–12975. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DKY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17028–17028. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

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concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BDI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14316–14316. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08JA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10843–10843. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0D6H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16522–16522. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05TC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7476–7476. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0BCC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14275–14275. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DL9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17039–17039. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04U0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6273–6273. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03WG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5111–5111. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0410: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5275–5275. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 064U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7878–7878. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 003T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 147–147. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 045G: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5435–5435. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 08K9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10877–10877. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04P2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6100–6100. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DLA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17040–17040. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DL1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17031–17031. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DL2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17032–17032. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 04SU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6232–6232. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0305: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3936–3936. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 07XL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10081–10081. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 05N0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7289–7289. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DKW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17026–17026. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 05F4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7013–7013. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07RT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9878–9878. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07SY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9918–9918. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07WN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10048–10048. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07W3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10028–10028. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DL7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17037–17037. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DL4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17034–17034. Stable Stacks tag for an expressly

allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 07WY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10058–10058. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0CXN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16213–16213. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07WU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10054–10054. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07X2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10062–10062. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07X1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10061–10061. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 05ZJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7693–7693. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DIZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16959–16959. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08IK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10818–10818. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0D0T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16323–16323. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0CXT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16218–16218. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original theorem 0DIQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16950–16950. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0CT7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16057–16057. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DL5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17035–17035. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 03YJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5186–5186. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 04X4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6382–6382. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07SK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9905–9905. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08K0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10868–10868. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08K1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10869–10869. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07Y4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10099–10099. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original proposition 07PX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9812–9812. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 07XJ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10079–10079. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0CXU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16219–16219. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Theorem 0.1 (Algebraicity of moduli of complexes on a proper morphism)**.** [1] *Let S be a scheme. Let $f : X \to B$ be morphism of algebraic spaces over S. Assume that f is proper, flat, and of finite presentation. Then $\mathcal{C}omplexes_{X/B}$ is an algebraic stack over S.*

1. Moduli of complexes on a proper morphism

The title and the material of this section are taken from [1]. Let S be a scheme and let $f : X \to B$ be a proper, flat, finitely presented morphism of algebraic spaces. We will prove that there is an algebraic stack

$$\mathcal{C}omplexes_{X/B}$$

parametrizing "families" of objects of D^b_{Coh} of the fibres with vanishing negative self-exts. More precisely a family is given by a relatively perfect object of the derived category of the total space; this somewhat technical notion is studied in More on Morphisms of Spaces, Section 0DKM.

Already if X is a proper algebraic space over a field k we obtain a very interesting algebraic stack. Namely, there is an embedding

$$\mathcal{C}oh_{X/k} \longrightarrow \mathcal{C}omplexes_{X/k}$$

since for any $\mathcal{O}$-module $\mathcal{F}$ (on any ringed topos) we have $\mathrm{Ext}^i_{\mathcal{O}}(\mathcal{F}, \mathcal{F}) = 0$ for $i < 0$. Although this certainly shows our stack is nonempty, the true motivation for the study of $\mathcal{C}omplexes_{X/k}$ is that there are often objects of the derived category $D^b_{Coh}(\mathcal{O}_X)$ with vanishing negative self-exts and nonvanishing cohomology sheaves in more than one degree. For example, X could be derived equivalent to another proper algebraic space Y over k, i.e., we have a k-linear equivalence

$$F : D^b_{Coh}(\mathcal{O}_Y) \longrightarrow D^b_{Coh}(\mathcal{O}_X)$$

There are cases where this happens and F is not given by an isomorphism between X and Y; for example in the case of an abelian variety and its dual. In this situation F induces an isomorphism of algebraic stacks

$$\mathcal{C}omplexes_{Y/k} \longrightarrow \mathcal{C}omplexes_{X/k}$$

(insert future reference here) and in particular the stack of coherent sheaves on Y maps into the stack of complexes on X. Turning this around, if we can understand

well enough the geometry of $\mathit{Complexes}_{X/k}$, then we can try to use this to study all possible derived equivalent Y.

Lemma 1.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is proper, flat, and of finite presentation. Let $K, E \in D(\mathcal{O}_X)$. Assume K is pseudo-coherent and E is Y-perfect (More on Morphisms of Spaces, Definition 0DKN). For a field k and a morphism $y : \mathrm{Spec}(k) \to Y$ denote K_y, E_y the pullback to the fibre X_y.*

(1) *There is an open $W \subset Y$ characterized by the property*
$$y \in |W| \Leftrightarrow \mathrm{Ext}^i_{\mathcal{O}_{X_y}}(K_y, E_y) = 0 \text{ for } i < 0.$$
(2) *For any morphism $V \to Y$ factoring through W we have*
$$\mathrm{Ext}^i_{\mathcal{O}_{X_V}}(K_V, E_V) = 0 \quad \text{for} \quad i < 0$$
where X_V is the base change of X and K_V and E_V are the derived pullbacks of K and E to X_V.
(3) *The functor $V \mapsto \mathrm{Hom}_{\mathcal{O}_{X_V}}(K_V, E_V)$ is a sheaf on $(\mathit{Spaces}/W)_{fppf}$ representable by an algebraic space affine and of finite presentation over W.*

Proof. For any morphism $V \to Y$ the complex K_V is pseudo-coherent (Cohomology on Sites, Lemma 08H4) and E_V is V-perfect (More on Morphisms of Spaces, Lemma 0DKT). Another observation is that given $y : \mathrm{Spec}(k) \to Y$ and a field extension k'/k with $y' : \mathrm{Spec}(k') \to Y$ the induced morphism, we have
$$\mathrm{Ext}^i_{\mathcal{O}_{X_{y'}}}(K_{y'}, E_{y'}) = \mathrm{Ext}^i_{\mathcal{O}_{X_y}}(K_y, E_y) \otimes_k k'$$
by Derived Categories of Schemes, Lemma 0AA7. Thus the vanishing in (1) is really a property of the induced point $y \in |Y|$. We will use these two observations without further mention in the proof.

Assume first Y is an affine scheme. Then we may apply More on Morphisms of Spaces, Lemma 0DKY and find a pseudo-coherent $L \in D(\mathcal{O}_Y)$ which "universally computes" $Rf_*R\mathcal{H}om(K, E)$ in the sense described in that lemma. Unwinding the definitions, we obtain for a point $y \in Y$ the equality
$$\mathrm{Ext}^i_{\kappa(y)}(L \otimes^{\mathbf{L}}_{\mathcal{O}_Y} \kappa(y), \kappa(y)) = \mathrm{Ext}^i_{\mathcal{O}_{X_y}}(K_y, E_y)$$
We conclude that
$$H^i(L \otimes^{\mathbf{L}}_{\mathcal{O}_Y} \kappa(y)) = 0 \text{ for } i > 0 \Leftrightarrow \mathrm{Ext}^i_{\mathcal{O}_{X_y}}(K_y, E_y) = 0 \text{ for } i < 0.$$
By Derived Categories of Schemes, Lemma 0BDI the set W of $y \in Y$ where this happens defines an open of Y. This open W then satisfies the requirement in (1) for all morphisms from spectra of fields, by the "universality" of L.

Let's go back to Y a general algebraic space. Choose an étale covering $\{V_i \to Y\}$ by affine schemes V_i. Then we see that the subset $W \subset |Y|$ pulls back to the corresponding subset $W_i \subset |V_i|$ for X_{V_i}, K_{V_i}, E_{V_i}. By the previous paragraph we find that W_i is open, hence W is open. This proves (1) in general. Moreover, parts (2) and (3) are entirely formulated in terms of the category Spaces/W and the restrictions X_W, K_W, E_W. This reduces us to the case $W = Y$.

Assume $W = Y$. We claim that for any algebraic space V over Y we have $Rf_{V,*}R\mathcal{H}om(K_V, E_V)$ has vanishing cohomology sheaves in degrees < 0. This will prove (2) because
$$\mathrm{Ext}^i_{\mathcal{O}_{X_V}}(K_V, E_V) = H^i(X_V, R\mathcal{H}om(K_V, E_V)) = H^i(V, Rf_{V,*}R\mathcal{H}om(K_V, E_V))$$

by Cohomology on Sites, Lemmas 08JA and 0D6H and the vanishing of the cohomology sheaves implies the cohomology group H^i is zero for $i < 0$ by Derived Categories, Lemma 05TC.

To prove the claim, we may work étale locally on V. In particular, we may assume Y is affine and $W = Y$. Let $L \in D(\mathcal{O}_Y)$ be as in the second paragraph of the proof. For an algebraic space V over Y denote L_V the derived pullback of L to V. (An important feature we will use is that L "works" for all algebraic spaces V over Y and not just affine V.) As $W = Y$ we have $H^i(L) = 0$ for $i > 0$ (use More on Algebra, Lemma 0BCC to go from fibres to stalks). Hence $H^i(L_V) = 0$ for $i > 0$. The property defining L is that

$$Rf_{V,*}R\mathcal{H}om(K_V, E_V) = R\mathcal{H}om(L_V, \mathcal{O}_V)$$

Since L_V sits in degrees ≤ 0, we conclude that $R\mathcal{H}om(L_V, \mathcal{O}_V)$ sits in degrees ≥ 0 thereby proving the claim. This finishes the proof of (2).

Assume $W = Y$ but make no assumptions on the algebraic space Y. Since we have (2), we see from Simplicial Spaces, Lemma 0DL9 that the functor F given by $F(V) = \mathrm{Hom}_{\mathcal{O}_{X_V}}(K_V, E_V)$ is a sheaf[1] on $(Spaces/Y)_{fppf}$. Thus to prove that F is an algebraic space and that $F \to Y$ is affine and of finite presentation, we may work étale locally on Y; see Bootstrap, Lemma 04U0 and Morphisms of Spaces, Lemmas 03WG and 0410. We conclude that it suffices to prove F is an affine algebraic space of finite presentation over Y when Y is an affine scheme. In this case we go back to our pseudo-coherent complex $L \in D(\mathcal{O}_Y)$. Since $H^i(L) = 0$ for $i > 0$, we can represent L by a complex of the form

$$\ldots \to \mathcal{O}_Y^{\oplus m_1} \to \mathcal{O}_Y^{\oplus m_0} \to 0 \to \ldots$$

with the last term in degree 0, see More on Algebra, Lemma 064U. Combining the two displayed formulas earlier in the proof we find that

$$F(V) = \mathrm{Ker}(\mathrm{Hom}_V(\mathcal{O}_V^{\oplus m_0}, \mathcal{O}_V) \to \mathrm{Hom}_V(\mathcal{O}_V^{\oplus m_1}, \mathcal{O}_V))$$

In other words, there is a fibre product diagram

$$\begin{array}{ccc} F & \longrightarrow & Y \\ \downarrow & & \downarrow 0 \\ \mathbf{A}_Y^{m_0} & \longrightarrow & \mathbf{A}_Y^{m_1} \end{array}$$

which proves what we want. □

Lemma 1.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is proper, flat, and of finite presentation. Let $E \in D(\mathcal{O}_X)$. Assume*

(1) *E is S-perfect (More on Morphisms of Spaces, Definition 0DKN), and*
(2) *for every point $s \in S$ we have*

$$\mathrm{Ext}^i_{\mathcal{O}_{X_s}}(E_s, E_s) = 0 \quad \text{for} \quad i < 0$$

where E_s is the pullback to the fibre X_s.

Then

(a) *(1) and (2) are preserved by arbitrary base change $V \to Y$,*
(b) *$\mathrm{Ext}^i_{\mathcal{O}_{X_V}}(E_V, E_V) = 0$ for $i < 0$ and all V over Y,*

[1] To check the sheaf property for a covering $\{V_i \to V\}_{i \in I}$ first consider the Čech fppf hypercovering $a : V_\bullet \to V$ with $V_n = \coprod_{i_0 \ldots i_n} V_{i_0} \times_V \ldots \times_V V_{i_n}$ and then set $U_\bullet = V_\bullet \times_{a,V} X_V$. Then $U_\bullet \to X_V$ is an fppf hypercovering to which we may apply Simplicial Spaces, Lemma 0DL9.

(c) $V \mapsto \mathrm{Hom}_{\mathcal{O}_{X_V}}(E_V, E_V)$ *is representable by an algebraic space affine and of finite presentation over* Y.

Here X_V *is the base change of* X *and* E_V *is the derived pullback of* E *to* X_V.

Proof. Immediate consequence of Lemma 1.1. □

Situation 1.3. Let S be a scheme. Let $f : X \to B$ be a morphism of algebraic spaces over S. Assume f is proper, flat, and of finite presentation. We denote $\mathit{Complexes}_{X/B}$ the category whose objects are triples (T, g, E) where

(1) T is a scheme over S,
(2) $g : T \to B$ is a morphism over S, and setting $X_T = T \times_{g,B} X$
(3) E is an object of $D(\mathcal{O}_{X_T})$ satisfying conditions (1) and (2) of Lemma 1.2.

A morphism $(T, g, E) \to (T', g', E')$ is given by a pair (h, φ) where

(1) $h : T \to T'$ is a morphism of schemes over B (i.e., $g' \circ h = g$), and
(2) $\varphi : L(h')^* E' \to E$ is an isomorphism of $D(\mathcal{O}_{X_T})$ where $h' : X_T \to X_{T'}$ is the base change of h.

Thus $\mathit{Complexes}_{X/B}$ is a category and the rule

$$p : \mathit{Complexes}_{X/B} \longrightarrow (\mathit{Sch}/S)_{fppf}, \quad (T, g, E) \longmapsto T$$

is a functor. For a scheme T over S we denote $\mathit{Complexes}_{X/B,T}$ the fibre category of p over T. These fibre categories are groupoids.

Lemma 1.4. *In Situation 1.3 the functor* $p : \mathit{Complexes}_{X/B} \longrightarrow (\mathit{Sch}/S)_{fppf}$ *is fibred in groupoids.*

Proof. We show that p is fibred in groupoids by checking conditions (1) and (2) of Categories, Definition 003T. Given an object (T', g', E') of $\mathit{Complexes}_{X/B}$ and a morphism $h : T \to T'$ of schemes over S we can set $g = h \circ g'$ and $E = L(h')^* E'$ where $h' : X_T \to X_{T'}$ is the base change of h. Then it is clear that we obtain a morphism $(T, g, E) \to (T', g', E')$ of $\mathit{Complexes}_{X/B}$ lying over h. This proves (1). For (2) suppose we are given morphisms

$$(h_1, \varphi_1) : (T_1, g_1, E_1) \to (T, g, E) \quad \text{and} \quad (h_2, \varphi_2) : (T_2, g_2, E_2) \to (T, g, E)$$

of $\mathit{Complexes}_{X/B}$ and a morphism $h : T_1 \to T_2$ such that $h_2 \circ h = h_1$. Then we can let φ be the composition

$$L(h')^* E_2 \xrightarrow{L(h')^* \varphi_2^{-1}} L(h')^* L(h_2)^* E = L(h_1)^* E \xrightarrow{\varphi_1} E_1$$

to obtain the morphism $(h, \varphi) : (T_1, g_1, E_1) \to (T_2, g_2, E_2)$ that witnesses the truth of condition (2). □

Lemma 1.5. *In Situation 1.3. Denote* $\mathcal{X} = \mathit{Complexes}_{X/B}$. *Then* $\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ *is representable by algebraic spaces.*

Proof. Consider two objects $x = (T, g, E)$ and $y = (T, g', E')$ of $\mathcal{X}$ over a scheme T. We have to show that $\mathit{Isom}_{\mathcal{X}}(x, y)$ is an algebraic space over T, see Algebraic Stacks, Lemma 045G. If for $h : T' \to T$ the restrictions $x|_{T'}$ and $y|_{T'}$ are isomorphic in the fibre category $\mathcal{X}_{T'}$, then $g \circ h = g' \circ h$. Hence there is a transformation of presheaves

$$\mathit{Isom}_{\mathcal{X}}(x, y) \longrightarrow \mathrm{Equalizer}(g, g')$$

Since the diagonal of B is representable (by schemes) this equalizer is a scheme. Thus we may replace T by this equalizer and E and E' by their pullbacks. Thus we may assume $g = g'$.

Assume $g = g'$. After replacing B by T and X by X_T we arrive at the following problem. Given $E, E' \in D(\mathcal{O}_X)$ satisfying conditions (1), (2) of Lemma 1.2 we have to show that $Isom(E, E')$ is an algebraic space. Here $Isom(E, E')$ is the functor

$$(Sch/B)^{opp} \to Sets, \quad T \mapsto \{\varphi : E_T \to E'_T \text{ isomorphism in } D(\mathcal{O}_{X_T})\}$$

where E_T and E'_T are the derived pullbacks of E and E' to X_T. Now, let $W \subset B$, resp. $W' \subset B$ be the open subspace of B associated to E, E', resp. to E', E by Lemma 1.1. Clearly, if there exists an isomorphism $E_T \to E'_T$ as in the definition of $Isom(E, E')$, then we see that $T \to B$ factors into both W and W' (because we have condition (1) for E and E' and we'll obviously have $E_t \cong E'_t$ so no nonzero maps $E_t[i] \to E_t$ or $E'_t[i] \to E_t$ over the fibre X_t for $i > 0$. Thus we may replace B by the open $W \cap W'$. In this case the functor $H = \mathcal{H}om(E, E')$

$$(Sch/B)^{opp} \to Sets, \quad T \mapsto \mathrm{Hom}_{\mathcal{O}_{X_T}}(E_T, E'_T)$$

is an algebraic space affine and of finite presentation over B by Lemma 1.1. The same is true for $H' = \mathcal{H}om(E', E)$, $I = \mathcal{H}om(E, E)$, and $I' = \mathcal{H}om(E', E')$. Therefore we can repeat the argument of the proof of Proposition 08K9 to see that

$$Isom(E, E') = (H' \times_B H) \times_{c, I \times_B I', \sigma} B$$

for some morphisms c and σ. Thus $Isom(E, E')$ is an algebraic space. □

Lemma 1.6. *In Situation 1.3 the functor $p : \mathcal{C}omplexes_{X/B} \longrightarrow (Sch/S)_{fppf}$ is a stack in groupoids.*

Proof. To prove that $\mathcal{C}omplexes_{X/B}$ is a stack in groupoids, we have to show that the presheaves *Isom* are sheaves and that descent data are effective. The statement on *Isom* follows from Lemma 1.5, see Algebraic Stacks, Lemma 045G. Let us prove the statement on descent data.

Suppose that $\{a_i : T_i \to T\}$ is an fppf covering of schemes over S. Let (ξ_i, φ_{ij}) be a descent datum for $\{T_i \to T\}$ with values in $\mathcal{C}omplexes_{X/B}$. For each i we can write $\xi_i = (T_i, g_i, E_i)$. Denote $\mathrm{pr}_0 : T_i \times_T T_j \to T_i$ and $\mathrm{pr}_1 : T_i \times_T T_j \to T_j$ the projections. The condition that $\xi_i|_{T_i \times_T T_j} \cong \xi_j|_{T_i \times_T T_j}$ implies in particular that $g_i \circ \mathrm{pr}_0 = g_j \circ \mathrm{pr}_1$. Thus there exists a unique morphism $g : T \to B$ such that $g_i = g \circ a_i$, see Descent on Spaces, Lemma 04P2. Denote $X_T = T \times_{g,B} X$. Set $X_i = X_{T_i} = T_i \times_{g_i, B} X = T_i \times_{a_i, T} X_T$ and

$$X_{ij} = X_{T_i} \times_{X_T} X_{T_j} = X_i \times_{X_T} X_j$$

with projections pr_i and pr_j to X_i and X_j. Observe that the pullback of (T_i, g_i, E_i) by $\mathrm{pr}_0 : T_i \times_T T_j \to T_i$ is given by $(T_i \times_T T_j, g_i \circ \mathrm{pr}_0, L\mathrm{pr}_i^* E_i)$. Hence a descent datum for $\{T_i \to T\}$ in $\mathcal{C}omplexes_{X/B}$ is given by the objects $(T_i, g \circ a_i, E_i)$ and for each pair i, j an isomorphism in $D\mathcal{O}_{X_{ij}}$)

$$\varphi_{ij} : L\mathrm{pr}_i^* E_i \longrightarrow L\mathrm{pr}_j^* E_j$$

satisfying the cocycle condition over the pullback of X to $T_i \times_T T_j \times_T T_k$. Using the vanishing of negative Exts provided by (b) of Lemma 1.2, we may apply Simplicial Spaces, Lemma 0DLA to obtain descent[2] for these complexes. In other words, we find there exists an object E in $D_{QCoh}(\mathcal{O}_{X_T})$ restricting to E_i on X_{T_i} compatible with φ_{ij}. Recall that being T-perfect signifies being pseudo-coherent and having locally finite tor dimension over $f^{-1}\mathcal{O}_T$. Thus E is T-perfect by an application of More on Morphisms of Spaces, Lemmas 0DL1 and 0DL2. Finally, we have to check

[2]To check this, first consider the Čech fppf hypercovering $a : T_\bullet \to T$ with $T_n = \coprod_{i_0 \ldots i_n} T_{i_0} \times_T \ldots \times_T T_{i_n}$ and then set $U_\bullet = T_\bullet \times_{a,T} X_T$. Then $U_\bullet \to X_T$ is an fppf hypercovering to which we may apply Simplicial Spaces, Lemma 0DLA.

condition (2) from Lemma 1.2 for E. This immediately follows from the description of the open W in Lemma 1.1 and the fact that (2) holds for E_i on X_{T_i}/T_i. □

Remark 1.7. In Situation 1.3 the rule $(T, g, E) \mapsto (T, g)$ defines a 1-morphism

$$\mathit{Complexes}_{X/B} \longrightarrow \mathcal{S}_B$$

of stacks in groupoids (see Lemma 1.6, Algebraic Stacks, Section 04SU, and Examples of Stacks, Section 0305). Let $B' \to B$ be a morphism of algebraic spaces over S. Let $\mathcal{S}_{B'} \to \mathcal{S}_B$ be the associated 1-morphism of stacks fibred in sets. Set $X' = X \times_B B'$. We obtain a stack in groupoids $\mathit{Complexes}_{X'/B'} \to (\mathit{Sch}/S)_{fppf}$ associated to the base change $f' : X' \to B'$. In this situation the diagram

$$\begin{array}{ccc} \mathit{Complexes}_{X'/B'} & \longrightarrow & \mathit{Complexes}_{X/B} \\ \downarrow & & \downarrow \\ \mathcal{S}_{B'} & \longrightarrow & \mathcal{S}_B \end{array} \quad \text{or in another notation} \quad \begin{array}{ccc} \mathit{Complexes}_{X'/B'} & \longrightarrow & \mathit{Complexes}_{X/B} \\ \downarrow & & \downarrow \\ \mathit{Sch}/B' & \longrightarrow & \mathit{Sch}/B \end{array}$$

is 2-fibre product square. This trivial remark will occasionally be useful to change the base algebraic space.

Lemma 1.8. *In Situation 1.3 assume that $B \to S$ is locally of finite presentation. Then $p : \mathit{Complexes}_{X/B} \to (\mathit{Sch}/S)_{fppf}$ is limit preserving (Artin's Axioms, Definition 07XL).*

Proof. Write $B(T)$ for the discrete category whose objects are the S-morphisms $T \to B$. Let $T = \lim T_i$ be a filtered limit of affine schemes over S. Assigning to an object (T, h, E) of $\mathit{Complexes}_{X/B,T}$ the object h of $B(T)$ gives us a commutative diagram of fibre categories

$$\begin{array}{ccc} \operatorname{colim} \mathit{Complexes}_{X/B,T_i} & \longrightarrow & \mathit{Complexes}_{X/B,T} \\ \downarrow & & \downarrow \\ \operatorname{colim} B(T_i) & \longrightarrow & B(T) \end{array}$$

We have to show the top horizontal arrow is an equivalence. Since we have assume that B is locally of finite presentation over S we see from Limits of Spaces, Remark 05N0 that the bottom horizontal arrow is an equivalence. This means that we may assume $T = \lim T_i$ be a filtered limit of affine schemes over B. Denote $g_i : T_i \to B$ and $g : T \to B$ the corresponding morphisms. Set $X_i = T_i \times_{g_i,B} X$ and $X_T = T \times_{g,B} X$. Observe that $X_T = \operatorname{colim} X_i$. By More on Morphisms of Spaces, Lemma 0DKW the category of T-perfect objects of $D(\mathcal{O}_{X_T})$ is the colimit of the categories of T_i-perfect objects of $D(\mathcal{O}_{X_{T_i}})$. Thus all we have to prove is that given an T_i-perfect object E_i of $D(\mathcal{O}_{X_{T_i}})$ such that the derived pullback E of E_i to X_T satisfies condition (2) of Lemma 1.2, then after increasing i we have that E_i satisfies condition (2) of Lemma 1.2. Let $W \subset |T_i|$ be the open constructed in Lemma 1.1 for E_i and E_i. By assumption on E we find that $T \to T_i$ factors through T. Hence there is an $i' \geq i$ such that $T_{i'} \to T_i$ factors through W, see Limits, Lemma 05F4 Then i' works by construction of W. □

Lemma 1.9. *In Situation 1.3. Let*

$$\begin{array}{ccc} Z & \longrightarrow & Z' \\ \downarrow & & \downarrow \\ Y & \longrightarrow & Y' \end{array}$$

be a pushout in the category of schemes over S where $Z \to Z'$ is a finite order thickening and $Z \to Y$ is affine, see More on Morphisms, Lemma 07RT. Then the functor on fibre categories

$$\mathit{Complexes}_{X/B,Y'} \longrightarrow \mathit{Complexes}_{X/B,Y} \times_{\mathit{Complexes}_{X/B,Z}} \mathit{Complexes}_{X/B,Z'}$$

is an equivalence.

Proof. Observe that the corresponding map

$$B(Y') \longrightarrow B(Y) \times_{B(Z)} B(Z')$$

is a bijection, see Pushouts of Spaces, Lemma 07SY. Thus using the commutative diagram

$$\begin{array}{ccc} \mathit{Complexes}_{X/B,Y'} & \longrightarrow & \mathit{Complexes}_{X/B,Y} \times_{\mathit{Complexes}_{X/B,Z}} \mathit{Complexes}_{X/B,Z'} \\ \downarrow & & \downarrow \\ B(Y') & \longrightarrow & B(Y) \times_{B(Z)} B(Z') \end{array}$$

we see that we may assume that Y' is a scheme over B'. By Remark 1.7 we may replace B by Y' and X by $X \times_B Y'$. Thus we may assume $B = Y'$.

Assume $B = Y'$. We first prove fully faithfulness of our functor. To do this, let ξ_1, ξ_2 be two objects of $\mathit{Complexes}_{X/B}$ over Y'. Then we have to show that

$$\mathit{Isom}(\xi_1, \xi_2)(Y') \longrightarrow \mathit{Isom}(\xi_1, \xi_2)(Y) \times_{\mathit{Isom}(\xi_1,\xi_2)(Z)} \mathit{Isom}(\xi_1, \xi_2)(Z')$$

is bijective. However, we already know that $\mathit{Isom}(\xi_1, \xi_2)$ is an algebraic space over $B = Y'$. Thus this bijectivity follows from Artin's Axioms, Lemma 07WN (or the aforementioned Pushouts of Spaces, Lemma 07SY).

Essential surjectivity. Let $(E_Y, E_{Z'}, \alpha)$ be a triple, where $E_Y \in D(\mathcal{O}_Y)$ and $E_{Z'} \in D(\mathcal{O}_{X_{Z'}})$ are objects such that $(Y, Y \to B, E_Y)$ is an object of $\mathit{Complexes}_{X/B}$ over Y, such that $(Z', Z' \to B, E_{Z'})$ is an object of $\mathit{Complexes}_{X/B}$ over Z', and $\alpha : L(X_Z \to X_Y)^* E_Y \to L(X_Z \to X_{Z'})^* E_{Z'}$ is an isomorphism in $D(\mathcal{O}_{Z'})$. That is to say

$$((Y, Y \to B, E_Y), (Z', Z' \to B, E_{Z'}), \alpha)$$

is an object of the target of the arrow of our lemma. Observe that the diagram

$$\begin{array}{ccc} X_Z & \longrightarrow & X_{Z'} \\ \downarrow & & \downarrow \\ X_Y & \longrightarrow & X_{Y'} \end{array}$$

is a pushout with $X_Z \to X_Y$ affine and $X_Z \to X_{Z'}$ a thickening (see Pushouts of Spaces, Lemma 07W3). Hence by Pushouts of Spaces, Lemma 0DL7 we find an object $E_{Y'} \in D(\mathcal{O}_{X_{Y'}})$ together with isomorphisms $L(X_Y \to X_{Y'})^* E_{Y'} \to E_Y$ and $L(X_{Z'} \to X_{Y'})^* E_{Y'} \to E_Z$ compatible with α. Clearly, if we show that $E_{Y'}$ is Y'-perfect, then we are done, because property (2) of Lemma 1.2 is a property on points (and Y and Y' have the same points). This follows from More on Morphisms of Spaces, Lemma 0DL4. □

Lemma 1.10. *In Situation 1.3 assume that S is a locally Noetherian scheme and $B \to S$ is locally of finite presentation. Let k be a finite type field over S and let $x_0 = (\mathrm{Spec}(k), g_0, E_0)$ be an object of $\mathcal{X} = \mathit{Complexes}_{X/B}$ over k. Then the spaces $T\mathcal{F}_{\mathcal{X},k,x_0}$ and $\mathit{Inf}(\mathcal{F}_{\mathcal{X},k,x_0})$ (Artin's Axioms, Section 07WY) are finite dimensional.*

Proof. Observe that by Lemma 1.9 our stack in groupoids $\mathcal{X}$ satisfies property (RS*) defined in Artin's Axioms, Section 0CXN. In particular $\mathcal{X}$ satisfies (RS). Hence all associated predeformation categories are deformation categories (Artin's Axioms, Lemma 07WU) and the statement makes sense.

In this paragraph we show that we can reduce to the case $B = \mathrm{Spec}(k)$. Set $X_0 = \mathrm{Spec}(k) \times_{g_0,B} X$ and denote $\mathcal{X}_0 = \mathit{Complexes}_{X_0/k}$. In Remark 1.7 we have seen that $\mathcal{X}_0$ is the 2-fibre product of $\mathcal{X}$ with $\mathrm{Spec}(k)$ over B as categories fibred in groupoids over $(\mathit{Sch}/S)_{fppf}$. Thus by Artin's Axioms, Lemma 07X2 we reduce to proving that B, $\mathrm{Spec}(k)$, and $\mathcal{X}_0$ have finite dimensional tangent spaces and infinitesimal automorphism spaces. The tangent space of B and $\mathrm{Spec}(k)$ are finite dimensional by Artin's Axioms, Lemma 07X1 and of course these have vanishing Inf. Thus it suffices to deal with $\mathcal{X}_0$.

Let $k[\epsilon]$ be the dual numbers over k. Let $\mathrm{Spec}(k[\epsilon]) \to B$ be the composition of $g_0 : \mathrm{Spec}(k) \to B$ and the morphism $\mathrm{Spec}(k[\epsilon]) \to \mathrm{Spec}(k)$ coming from the inclusion $k \to k[\epsilon]$. Set $X_0 = \mathrm{Spec}(k) \times_B X$ and $X_\epsilon = \mathrm{Spec}(k[\epsilon]) \times_B X$. Observe that X_ϵ is a first order thickening of X_0 flat over the first order thickening $\mathrm{Spec}(k) \to \mathrm{Spec}(k[\epsilon])$. Observe that X_0 and X_ϵ give rise to canonically equivalent small étale topoi, see More on Morphisms of Spaces, Section 05ZJ. By More on Morphisms of Spaces, Lemma 0DL4 we see that $T\mathcal{F}_{\mathcal{X}_0,k,x_0}$ is the set of isomorphism classes of lifts of E_0 to X_ϵ in the sense of Deformation Theory, Lemma 0DIZ. We conclude that

$$T\mathcal{F}_{\mathcal{X}_0,k,x_0} = \mathrm{Ext}^1_{\mathcal{O}_{X_0}}(E_0, E_0)$$

Here we have used the identification $\epsilon k[\epsilon] \cong k$ of $k[\epsilon]$-modules. Using Deformation Theory, Lemma 0DIZ once more we see that there is a surjection

$$\mathrm{Inf}(\mathcal{F}_{\mathcal{X},k,x_0}) \leftarrow \mathrm{Ext}^0_{\mathcal{O}_{X_0}}(E_0, E_0)$$

of k-vector spaces. As E_0 is pseudo-coherent it lies in $D^-_{Coh}(\mathcal{O}_{X_0})$ by Derived Categories of Spaces, Lemma 08IK. Since E_0 locally has finite tor dimension and X_0 is quasi-compact we see $E_0 \in D^b_{Coh}(\mathcal{O}_{X_0})$. Thus the Exts above are finite dimensional k-vector spaces by Derived Categories of Spaces, Lemma 0D0T. □

Lemma 1.11. *In Situation 1.3 assume $B = S$ is locally Noetherian. Then strong formal effectiveness in the sense of Artin's Axioms, Remark 0CXT holds for $p : \mathit{Complexes}_{X/S} \to (\mathit{Sch}/S)_{fppf}$.*

Proof. Let (R_n) be an inverse system of S-algebras with surjective transition maps whose kernels are locally nilpotent. Set $R = \lim R_n$. Let (ξ_n) be a system of objects of $\mathit{Complexes}_{X/B}$ lying over $(\mathrm{Spec}(R_n))$. We have to show (ξ_n) is effective, i.e., there exists an object ξ of $\mathit{Complexes}_{X/B}$ lying over $\mathrm{Spec}(R)$.

Write $X_R = \mathrm{Spec}(R) \times_S X$ and $X_n = \mathrm{Spec}(R_n) \times_S X$. Of course X_n is the base change of X_R by $R \to R_n$. Since $S = B$, we see that ξ_n corresponds simply to an R_n-perfect object $E_n \in D(\mathcal{O}_{X_n})$ satisfying condition (2) of Lemma 1.2. In particular E_n is pseudo-coherent. The isomorphisms $\xi_{n+1}|_{\mathrm{Spec}(R_n)} \cong \xi_n$ correspond to isomorphisms $L(X_n \to X_{n+1})^* E_{n+1} \to E_n$. Therefore by Flatness on Spaces, Theorem 0DIQ we find a pseudo-coherent object E of $D(\mathcal{O}_{X_R})$ with E_n equal to the derived pullback of E for all n compatible with the transition isomorphisms.

Observe that $(R, \mathrm{Ker}(R \to R_1))$ is a henselian pair, see More on Algebra, Lemma 0CT7. In particular, $\mathrm{Ker}(R \to R_1)$ is contained in the Jacobson radical of R. Then we may apply More on Morphisms of Spaces, Lemma 0DL5 to see that E is R-perfect.

Finally, we have to check condition (2) of Lemma 1.2. By Lemma 1.1 the set of points t of $\mathrm{Spec}(R)$ where the negative self-exts of E_t vanish is an open. Since this

condition is true in $V(\mathrm{Ker}(R \to R_1))$ and since $\mathrm{Ker}(R \to R_1)$ is contained in the Jacobson radical of R we conclude it holds for all points. □

Proof. Set $\mathcal{X} = \mathit{Complexes}_{X/B}$. We have seen that $\mathcal{X}$ is a stack in groupoids over $(\mathit{Sch}/S)_{fppf}$ with diagonal representable by algebraic spaces (Lemmas 1.6 and 1.5). Hence it suffices to find a scheme W and a surjective and smooth morphism $W \to \mathcal{X}$.

Let B' be a scheme and let $B' \to B$ be a surjective étale morphism. Set $X' = B' \times_B X$ and denote $f' : X' \to B'$ the projection. Then $\mathcal{X}' = \mathit{Complexes}_{X'/B'}$ is equal to the 2-fibre product of $\mathcal{X}$ with the category fibred in sets associated to B' over the category fibred in sets associated to B (Remark 1.7). By the material in Algebraic Stacks, Section 03YJ the morphism $\mathcal{X}' \to \mathcal{X}$ is surjective and étale. Hence it suffices to prove the result for $\mathcal{X}'$. In other words, we may assume B is a scheme.

Assume B is a scheme. In this case we may replace S by B, see Algebraic Stacks, Section 04X4. Thus we may assume $S = B$.

Assume $S = B$. Choose an affine open covering $S = \bigcup U_i$. Denote $\mathcal{X}_i$ the restriction of $\mathcal{X}$ to $(\mathit{Sch}/U_i)_{fppf}$. If we can find schemes W_i over U_i and surjective smooth morphisms $W_i \to \mathcal{X}_i$, then we set $W = \coprod W_i$ and we obtain a surjective smooth morphism $W \to \mathcal{X}$. Thus we may assume $S = B$ is affine.

Assume $S = B$ is affine, say $S = \mathrm{Spec}(\Lambda)$. Write $\Lambda = \mathrm{colim}\, \Lambda_i$ as a filtered colimit with each Λ_i of finite type over $\mathbf{Z}$. For some i we can find a morphism of algebraic spaces $X_i \to \mathrm{Spec}(\Lambda_i)$ which is proper, flat, of finite presentation and whose base change to Λ is X. See Limits of Spaces, Lemmas 07SK, 08K0, and 08K1. If we show that $\mathit{Complexes}_{X_i/\mathrm{Spec}(\Lambda_i)}$ is an algebraic stack, then it follows by base change (Remark 1.7 and Algebraic Stacks, Section 04X4) that $\mathcal{X}$ is an algebraic stack. Thus we may assume that Λ is a finite type $\mathbf{Z}$-algebra.

Assume $S = B = \mathrm{Spec}(\Lambda)$ is affine of finite type over $\mathbf{Z}$. In this case we will verify conditions (1), (2), (3), (4), and (5) of Artin's Axioms, Lemma 07Y4 to conclude that $\mathcal{X}$ is an algebraic stack. Note that Λ is a G-ring, see More on Algebra, Proposition 07PX. Hence all local rings of S are G-rings. Thus (5) holds. To check (2) we have to verify axioms [-1], [0], [1], [2], and [3] of Artin's Axioms, Section 07XJ. We omit the verification of [-1] and axioms [0], [1], [2], [3] correspond respectively to Lemmas 1.6, 1.8, 1.9, 1.10. Condition (3) follows from Lemma 1.11. Condition (1) is Lemma 1.5.

It remains to show condition (4) which is openness of versality. To see this we will use Artin's Axioms, Lemma 0CXU. We have already seen that $\mathcal{X}$ has diagonal representable by algebraic spaces, has (RS*), and is limit preserving (see lemmas used above). Hence we only need to see that $\mathcal{X}$ satisfies the strong formal effectiveness formulated in Artin's Axioms, Lemma 0CXU. This follows from Lemma 1.11 and the proof is complete. □

References

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